Notes: Roberts (AJPS, 1990)

Political Institutions, Policy Expectations, and the 1980 Election: A Financial Market Perspective

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1 One-sentence summary

Roberts uses stock-market reactions of defense contractors around the 1980 U.S. election to infer how investors separately valued the expected defense-spending effects of a Reagan presidency and a Republican Senate.

2 Big picture

2.1 Research question

The paper asks whether financial markets can identify the separate policy consequences of two simultaneous political shocks: Ronald Reagan winning the presidency and Republicans gaining control of the Senate in 1980.

2.2 Empirical idea

If a political event changes expected defense spending, the expected profits of firms exposed to Department of Defense (DoD) contracts should change immediately. A defense-industry portfolio therefore becomes a market-based measure of expected policy change.

2.3 Main conclusion

Both a rising probability of a Reagan victory and the realization of Republican Senate control increase the returns of a defense-sensitive portfolio. The Senate shock is especially large, suggesting that investors viewed Senate control as an important marginal input into future defense policy.

2.4 Why this matters

The study treats securities prices as a quantitative measure of expected policy influence. Instead of relying on qualitative claims about presidents or Congress, it extracts institutional expectations from prices of politically exposed firms.

3 Incremental contribution of the paper

3.1 A financial-market measure of political institutions

The paper introduces a capital-market strategy for measuring institutional influence. It uses security prices to infer the present value of expected policy changes associated with different branches of government.

3.2 Separating presidential and Senate shocks

The 1980 election is unusual because party control changes in both the White House and the Senate. The paper exploits different timing and probability structures to separate the Reagan-presidency shock from the Republican-Senate shock.

3.3 Sector-specific political sensitivity

Rather than studying the whole stock market, the paper constructs a defense portfolio weighted by the share of each firm's sales coming from the Department of Defense. This creates a portfolio that should be especially sensitive to defense-policy expectations.

3.4 Combining event-study logic with probability data

The paper does not only use election-day dummies. It uses betting odds from Ladbrokes to construct daily changes in the probability of a Reagan victory, allowing expected policy changes to be measured continuously before Election Day.

3.5 Institutional interpretation beyond partisan control

The paper emphasizes that Reagan's election and the Republican Senate victory do not have identical institutional meanings. The presidential coefficient reflects a Republican president facing a Democratic Senate baseline; the Senate coefficient reflects the marginal value of unified Republican control after Reagan's victory was nearly certain.

4 Data and measurement

4.1 Defense-industry portfolio

The dependent variable is the daily return on a portfolio of 58 defense-related firms. Firms are chosen from the Department of Defense list of the top 100 defense contractors in fiscal year 1981, subject to having Compustat sales data and CRSP daily returns.

4.2 Portfolio weights

The portfolio weights are designed to measure political sensitivity to defense spending. For firm i,

$$W_i \propto \frac{DSales_i}{TSales_i},$$

where $DSales_i$ is fiscal-year 1981 sales to the Department of Defense and $TSales_i$ is total sales in 1980. The portfolio return is

$$R_{pt} = \sum_i W_i R_{it}, \qquad \sum_i W_i = 1.$$

4.3 Market return control

The model controls for the CRSP equal-weighted market return, R_{mt}, to separate politically specific defense-sector reactions from broad market movements.

4.4 Presidential probability variable

The presidential variable, P_t, is the daily change in the probability that Ronald Reagan wins the 1980 presidential election. It is constructed from Ladbrokes betting odds. The probability series evolves during the campaign and resolves after Election Day.

4.5 Senate probability variable

The Senate variable, S_t, equals one on the first trading day after the election and zero otherwise. This reflects the assumption that the probability of Republican Senate control was effectively resolved once all Senate results were known after the election.

4.6 Why the two political variables differ

The Reagan probability varies gradually before Election Day, while the Senate variable is concentrated in one post-election observation. This difference lets the model separately estimate the value of a Republican presidency and the value of a Republican Senate.

5 Empirical specification

5.1 Modified market model

The main regression is

$$R_{pt} = \alpha + \beta R_{mt} + \delta_1 P_t + \delta_2 S_t + v_t.$$

Here δ_1 measures the defense-portfolio return associated with a change in Reagan's election probability, while δ_2 measures the effect of the Republican Senate victory.

5.2 Cross-sectional validation

Before forming the final weighted portfolio, the paper checks whether firm-level reactions to the political variables are increasing in DoD-sales exposure. For each firm,

$$R_{it} = \alpha_i + \beta_i R_{mt} + \gamma_{1i} P_t + \gamma_{2i} S_t + v_{it}.$$

The estimated γ's are then related to the portfolio weight W_i. The significant relationship validates the DoD-sales weights as political-sensitivity weights.

5.3 Interpretation of coefficients

A coefficient of 0.01 in the portfolio regression corresponds to an abnormal return of one percent on the defense portfolio. Since the portfolio is built from firms with defense exposure, the coefficients are interpreted as the capitalized value of expected changes in defense-related corporate profitability.

6 Identification and empirical design

6.1 What is identified

The paper identifies market expectations about policy influence, not actual causal policy effects. The coefficient estimates measure how investors expected defense spending and defense-firm profits to change when electoral probabilities changed.

6.2 Why defense contractors provide the signal

Defense contractors are the politically exposed assets. If investors expect Republican control to raise defense expenditures, firms with large DoD exposure should gain relative to the market.

6.3 Timing and market efficiency

The design assumes that capital markets rapidly incorporate public information. Changes in election probabilities and election outcomes should be reflected in stock prices when the information becomes available.

6.4 Separating the president from the Senate

The key design challenge is that the presidency and Senate both change in 1980. The paper separates them because Reagan's probability changes over many campaign days, while the Senate switch is revealed essentially on the first trading day after the election.

6.5 Confounding-event concern

Any confounding event would need to explain why firms with higher DoD exposure react more strongly. The cross-sectional validation using DoD sales reduces the plausibility that a nonpolitical event drives the results.

6.6 Nonlinear election-reaction concern

The author discusses the possibility that the post-election jump captures a nonlinear presidential landslide effect rather than the Senate effect. The paper argues that pre-election probability movements and model checks reduce this concern, but it remains a limitation.

6.7 Main limitation

The method applies best to policy areas with clear publicly traded beneficiaries. It can measure expected policy influence for defense spending, but not all policy domains have equally clean traded proxies.

7 Tables and figures

7.1 Table 1: Summary Statistics: Defense Portfolio; Figure 1: The 1980 Presidential Election

Read:

- Table 1 describes the defense portfolio of 58 firms. Mean total sales are about \$13.5 billion, and mean DoD sales are 17.4 percent of total sales.
- The DoD exposure distribution is highly skewed. Grumman has 89.25 percent of sales to DoD, while DuPont has only 0.58 percent.
- Figure 1 plots the market-implied probability of Reagan winning over the 60 trading days before the election.
- Reagan's probability declines through much of the campaign, rises late, and then resolves to one at the election.

Economic message: The table constructs the political-exposure portfolio, while the figure constructs the pre-election political probability shock. Together, they create the dependent variable and the main presidential explanatory variable.

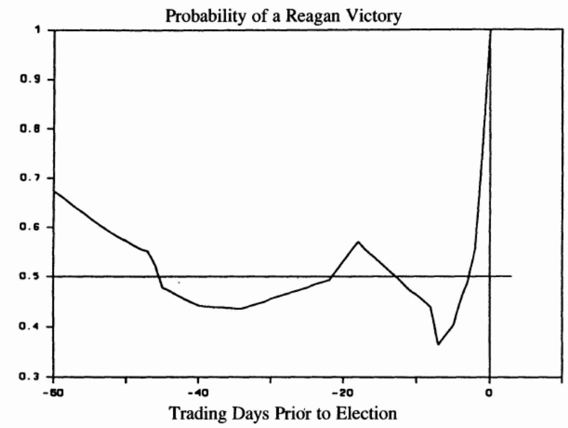
Detailed interpretation: The portfolio-weighting strategy matters because not all defense contractors are equally sensitive to defense policy. Exxon is large but barely defense-dependent, while Grumman is far more exposed. The figure shows why probability data are useful: the presidential election is not just a one-day event but a sequence of changing beliefs.

Table 1. Summary Statistics: Defense Portfolio

Summary				
Total sales (millions $):				
Mean: $13,520.30				
Quantiles:	100%	Max	$108,108	(Exxon Corp.)
	75%	Q3	$11,686	
	50%	Med	$5,603	
	25%	Q1	$3,083	
	0%	Min	$364	(Sanders Assoc. Inc.)
Percentage total sales to DoD (FY81):				
Mean: 17.4%				
Quantiles:	100%	Max	89.25%	(Grumman Corp.)
	75%	Q3	27.45%	
	50%	Med	10.68%	
	25%	Q1	3.50%	
	0%	Min	.58%	(DuPont E I de Nemours & Co.)

(a) Table 1: Summary Statistics: Defense Portfolio.

Figure 1. The 1980 Presidential Election



(b) Figure 1: The 1980 Presidential Election.

7.2 Table 2: Estimation Results for Modified Market Model; Appendix A: Defense Industry Portfolio

Read:

- Table 2 reports a market beta of 1.0708 and an R^2 of 0.8752, so the market model explains most daily variation in the defense portfolio.
- The Reagan probability coefficient is 0.0147 with a t-ratio of 2.131.
- The Senate probability coefficient is 0.0725 with a t-ratio of 3.664.
- Appendix A lists the companies in the defense industry portfolio ranked by portfolio weight.

Economic message: The market expected both a Reagan victory and Republican Senate control to increase defense-related firm values. The larger Senate coefficient suggests that investors saw Senate control as a major marginal force for defense-spending expansion.

Detailed interpretation: The coefficients should not be read as pure institutional dominance. The presidential coefficient is measured against a baseline in which the Senate was expected to remain Democratic. The Senate coefficient is measured after Reagan's victory had become nearly certain. The right interpretation is institutional complementarity: the Senate result magnifies the expected ability of the Reagan administration to implement defense policy.

Table 2. Estimation Results for Modified Market Model

$$R_{pt} = \alpha + \beta R_{mt} + \delta_1 P_t + \delta_2 S_t + v_t$$

Variable	Coefficient	Beta	T-ratio
Intercept (α)	.0005		0.824
Market index (β)	1.0708		17.375
Change in Reagan probability (δ_1)	.0147	.0913	2.131
Change in Senate probability (δ_2)	.0725	.2483	3.664
$R^2 = .8752$			
$N = 63$			

(a) Table 2: Estimation Results for Modified Market Model.

APPENDIX A

Defense Industry Portfolio (Ranked by Portfolio Weight)

Company	Total Sales (M$)	DoD Sales (M$)	Portfolio Weight
1. Grumman Corp.	1,916	1,710	0.0867
2. General Dynamics Corp.	5,063	3,402	0.0652
3. Todd Shipyards Corp.	716	472	0.0640
4. McDonnell Douglas Corp.	7,385	4,409	0.0580
5. Sanders Assoc. Inc.	364	190	0.0507
6. Lockheed Corp.	5,176	2,656	0.0498

(b) Appendix A: Defense Industry Portfolio, top-weighted firms.

7.3 Appendix A: Defense Industry Portfolio, continued

Read:

- Appendix A lists all 58 firms, their total sales, DoD sales, and portfolio weights.
- Top portfolio weights go to firms with high DoD-sales dependence, not necessarily the largest firms by total sales.
- The portfolio is therefore a political-exposure portfolio rather than a simple industry-size portfolio.

Economic message: The appendix is important because the paper's empirical strategy relies on portfolio construction. The signal comes from firms whose profits are expected to be most sensitive to defense spending.

Detailed interpretation: The high weight on Grumman, General Dynamics, Todd Shipyards, McDonnell Douglas, Sanders, and Lockheed reflects defense intensity. The low weights on Exxon, General Motors, IBM, and DuPont reflect their small DoD shares despite large total sales.

APPENDIX A (*continued*)

Company	Total Sales (M$)	DoD Sales (M$)	Portfolio Weight
7. E Sys Inc.	572	275	0.0467
8. Martin Marietta Corp.	3,294	1,286	0.0379
9. Fairchild Inds. Inc.	1,339	457	0.0332
10. Raytheon Co.	5,636	1,825	0.0314
11. Northrop Corp.	1,991	623	0.0304
12. FMC Corp.	3,367	1,052	0.0303
13. Litton Inds. Inc.	4,936	1,384	0.0272
14. United Technologies Corp.	13,668	3,775	0.0268
15. Boeing Co.	9,788	2,682	0.0266
16. Pacific Res. Inc. Hawaii	1,080	243	0.0218
17. Harris Corp. Del.	1,552	263	0.0165
18. Sperry Corp.	5,571	928	0.0161
19. Rockwell Intl. Corp.	7,040	1,125	0.0155
20. Honeywell Inc.	5,351	838	0.0152
21. GATX Corp.	1,191	185	0.0151
22. Texas Instrs. Inc.	4,206	625	0.0144
23. North American Philips Corp.	3,030	406	0.0130
24. Chrysler Corp.	10,822	1,414	0.0127
25. American Mtrs. Corp.	2,589	326	0.0122
26. Ex Cello Corp.	1,124	139	0.0120
27. Westinghouse Elec. Corp.	9,368	1,124	0.0116
28. General Elec. Co.	27,240	3,018	0.0107
29. RCA Corp.	8,005	876	0.0106
30. Coastal Corp.	5,910	616	0.0101
31. Hercules Inc.	2,718	281	0.0100
32. TRW Inc.	5,285	516	0.0094
33. Lear Siegler Inc.	1,531	144	0.0091
34. Tenneco Inc.	15,462	1,151	0.0072
35. LTV Corp.	7,511	548	0.0070
36. Emerson Elec. Co.	3,429	237	0.0067
37. Motorola Inc.	3,336	199	0.0058
38. Burroughs Corp.	3,318	176	0.0051
39. Control Data Corp. Del.	3,101	161	0.0050
40. Phibro Salomon Inc.	25,098	1,223	0.0047
41. Signal Cos. Inc.	5,343	238	0.0043

(a) Appendix A: Defense portfolio, middle-ranked firms.

APPENDIX A (*continued*)

Company	Total Sales (M$)	DoD Sales (M$)	Portfolio Weight
48. Atlantic Richfield Co.	27,797	547	0.0019
49. Ashland Oil Inc.	9,262	182	0.0019
50. Xerox Corp.	8,691	132	0.0014
51. Ford Mtr. Co. Del.	38,247	543	0.0013
52. Eastman Kodak Co.	10,337	125	0.0011
53. American Tel. & Teleg. Co.	58,214	694	0.0011
54. Exxon Corp.	108,108	1,152	0.0010
55. General Mtrs. Corp.	62,699	621	0.0009
56. Texaco Inc.	57,628	379	0.0006
57. Mobil Corp.	64,448	385	0.0005
58. DuPont E. I. de Nemours & Co.	2,281	132	0.0005

Note: Total sales data acquired from the COMPUSTAT data base

(b) Appendix A: Defense portfolio, low-weight firms and notes.

7.4 Notes-created table: Main empirical objects and interpretations

Object	Definition	Interpretation
Defense portfolio return	Weighted return of 58 DoD-exposed firms	Asset-price signal of expected defense-policy effects
P_t	Change in probability Reagan wins	Presidential election expectation shock
S_t	Post-election Senate Republican-control indicator	Senate election resolution shock
δ_1	Coefficient on P_t	Expected effect of Reagan presidency with Democratic Senate baseline
δ_2	Coefficient on S_t	Marginal value of Republican Senate control after Reagan victory
DoD weight	Function of DoD sales over total sales	Cross-sectional political sensitivity measure

Notes-created summary table: empirical objects in Roberts (1990). This is not an original table from the paper; it summarizes how the variables map to the paper's institutional interpretation.

8 Short summary

- The paper uses asset prices to study political institutions.
- The empirical setting is the 1980 election, which changed control of both the presidency and the Senate.
- The policy focus is defense spending.
- The dependent variable is the return on a defense-industry portfolio weighted by DoD-sales exposure.
- The presidential variable is the daily change in the betting-market probability of a Reagan victory.
- The Senate variable is a post-election indicator for the realization of Republican control.
- Both variables positively predict defense-portfolio returns.
- The Senate coefficient is substantially larger than the presidential coefficient.
- The paper interprets the results as evidence that the Senate mattered greatly for expected defense policy.
- The broader methodological contribution is using financial markets to measure expected policy effects.

9 Conclusion

9.1 Core conclusion

Roberts shows that financial markets can be used to infer the expected policy consequences of political events. In the 1980 election, markets expected both Reagan's election and Republican Senate control to increase defense-firm profitability, but the Senate shock produced a particularly large defense-portfolio response.

9.2 Economic intuition

Defense contractors are valued partly based on expected future government spending. If an election outcome changes expected defense budgets, the present value of expected contractor cash flows changes immediately. Stock returns can therefore be used as a high-frequency measure of political expectations.

9.3 Institutional interpretation

The results support the idea that Congress matters for policy expectations, especially in budgetary domains such as defense. The large Senate coefficient suggests that investors believed Republican Senate control would make Reagan's defense agenda much easier to implement.

9.4 Incremental contribution revisited

The paper's contribution is to make institutional power observable in market data. It treats prices of politically sensitive assets as a measurement device for public-policy expectations, adding a financial-market perspective to the study of separation of powers.

9.5 Limitations

The identification depends on efficient capital markets, accurate interpretation of election timing, and absence of confounding events with the same cross-sectional DoD-sales pattern. The Senate variable is concentrated in one trading day, so it is more vulnerable to nonlinear election-day reactions than the presidential probability variable.

9.6 Takeaway

Political institutions affect policy expectations, and those expectations can be capitalized into asset prices. In 1980, markets viewed the Republican Senate victory as a major complement to Reagan's presidency in shaping expected defense spending.